AI Agentic Workflow for Causal Inference:

Introduction to ./causal_inference_ai_agentic_workflow

⚠️ **Safe Harbor Statement**

This is an experimental AI-assisted workflow for causal inference. AI outputs can be plausible but incorrect — all methods and results must be critically reviewed. This tool requires solid causal inference understanding and is not a substitute for rigorous methodological knowledge. Validate independently before drawing conclusions.

A **design-first framework** for estimating causal effects, powered by four specialized AI agents in VS Code.

This workflow helps you move from a causal question to a reproducible analysis pipeline while keeping identification assumptions, diagnostics, and robustness checks explicit.

Agent Roles

Planner
Formulates the causal question, drafts the DAG, checks identification assumptions, and produces an approved analysis plan.

Implementer
Builds the estimation pipeline from data preparation through estimation, diagnostics, robustness checks, and reporting. It is instructed to stop rather than emit results when key diagnostics fail.

Reviewer
Audits both the causal design and the code, then routes findings back to the appropriate agent until the analysis is clean.

Orchestrator
An optional single entry point that coordinates the Planner, Implementer, and Reviewer automatically.

Skills Included in the Workflow

causal-inference
An adapted version of the causal inference skill by Alexandre Andorra from *Learning Bayesian Statistics*. This version has been adjusted to make the workflow more broadly accessible, moving beyond a primarily Bayesian analytical framing while preserving the

core emphasis on causal design, identification, DAG-based reasoning, estimation strategy, diagnostics, and robustness checks.

docx
A document-generation skill from Anthropic used to create Word documents describing the analysis. This enables the workflow to produce structured, shareable reports that document the causal question, assumptions, analysis plan, diagnostics, robustness checks, and interpretation of results.

How to Use the Workflow

1. Fill in `project.yaml` with your causal question, treatment, outcomes, and covariates.
2. Place your raw data in the raw/ folder.
3. Ask the Planner to prepare the analysis plan.
4. Answer the Planner's clarifying questions.
5. Hand the approved plan to the Implementer.
6. Route the completed analysis to the Reviewer.
7. Iterate until the workflow passes review.

Current Configuration

This repository is currently pre-configured for the **manager training causal impact evaluation** use case. To adapt it to your own analysis, update `project.yaml `with your study details and place your data in raw/. The agent roles, workflow rules, and guardrails remain the same regardless of the specific causal question.

Platform Note

Some paths and workflow conventions are currently set up for **GitHub Copilot implementation in VS Code**. They can be easily adjusted for other agentic coding platforms such as **Cursor**, **Claude Code**, or similar AI-assisted development environments.